Oracle SQL Developer

Download, install, configure, and connect to an Oracle Database

Audience	Beginners and workshop participants
Platforms	Windows and macOS
Official source	Oracle SQL Developer Downloads

What this guide covers

Finding the correct Oracle download, choosing the right package for your operating system, launching SQL Developer, creating a database connection, and testing it with a simple SQL query.

> **Important:** SQL Developer is a database client. It does not install or create an Oracle Database by itself. You need connection details and a database account from your instructor or database administrator.

1. Before you begin

Have these items ready before installing or connecting:

• A supported Windows or macOS computer with permission to download and extract files.

• Internet access to Oracle's official download page.

• Database connection information: username, password, hostname/service name or TNS alias, and network access.

• For Oracle Autonomous Database using wallet-based connectivity: the wallet ZIP file provided by the database administrator or instructor, plus any required credentials.

Current download choice

At the time this guide was prepared, Oracle's download page lists SQL Developer 26.2. Choose the package matching your operating system and processor. The available packages and labels can change, so use the official page rather than third-party download sites.

Open Oracle SQL Developer Downloads

2. Download SQL Developer from Oracle

1. Open Oracle's official page. Go to Oracle SQL Developer Downloads.

2. Review the platform choices. Find the package for your operating system. Read the package label carefully, especially whether it includes the Java Development Kit (JDK).

3. Accept the license terms. If prompted, review and accept Oracle's license agreement before downloading.

4. Download the archive. Save the ZIP or platform-specific archive to a folder you can find, such as Downloads.

5. Keep the archive intact until extraction. Do not try to run SQL Developer from inside the compressed ZIP file.

Which package should I choose?

Operating system	Recommended selection	Java requirement
Windows 64-bit	Windows x64 package with JDK included, if offered	Usually no separate JDK installation needed
Mac with Apple Silicon (M-series)	macOS ARM64 / Apple Silicon package	Choose the package that includes JDK when offered
Mac with Intel processor	macOS Intel / x64 package	Choose the package that includes JDK when offered
Linux	Linux package matching your distribution and architecture	Follow Oracle's platform-specific instructions

Tip: The package that includes JDK is generally the simplest option for a new learner because it reduces Java compatibility and configuration issues. Always confirm the exact package notes on Oracle's download page.

3. Install and launch on Windows

1. Open File Explorer and locate the downloaded ZIP archive.

2. Right-click the ZIP file and choose **Extract All…**. Extract it to a dedicated folder, for example `C:\Tools\OracleSQLDeveloper`.

3. Open the extracted folder. Depending on the package, the executable may be in the top-level folder or inside a folder named `sqldeveloper`.

4. Double-click `sqldeveloper.exe` to launch the application.

5. If SQL Developer asks for a Java/JDK location, this usually means you downloaded a package without an included JDK or the JDK was not detected. Install the JDK version required by the release and select the JDK home folder—not the `bin` subfolder—when prompted.

6. After SQL Developer opens, optionally pin it to Start or create a shortcut to the executable.

4. Install and launch on macOS

1. On Oracle's download page, choose the package for your Mac's processor: Apple Silicon/ARM64 for M-series Macs, or Intel/x64 for Intel Macs.

2. Open Finder → Downloads and locate the downloaded archive.

3. Double-click the archive to extract it. If Oracle provides a native macOS application package, move the extracted SQL Developer app to the Applications folder.

4. Open SQL Developer from Applications. If macOS blocks the app, confirm that it came from Oracle's official website and follow your organization's security policy; do not bypass security prompts for an untrusted download.

5. If you downloaded a generic archive rather than a native app, follow the launch instructions included with that exact package. Some packages are launched using the supplied script from Terminal.

> Avoid replacing an existing SQL Developer folder during extraction. Use a new folder for a new version so you can retain the previous version until the new one is confirmed to work.

5. Create a database connection

The following steps apply to a typical Oracle Database connection. Your screen labels may vary slightly by release.

1. Open SQL Developer. In the left-side **Connections** panel, click the **+** icon or right-click Connections and choose **New Connection**.

2. Enter a recognizable connection name, such as `HR_Practice`.

3. Enter the database username and password supplied by your instructor or administrator.

4. Select the connection type. For a normal host/port/service connection, choose **Basic**. Enter the hostname, port (commonly `1521`), and service name. Do not assume the service name is the same as the database SID.

5. For a managed Oracle Autonomous Database connection, select the connection option supported by your SQL Developer version and the database's instructions. If wallet-based connectivity is required, use the supplied wallet and TNS alias as instructed.

6. Click **Test**. If the status reports **Success**, click **Connect**.

Example: typical Basic connection

Field	Example value	Notes
Connection Name	HR_Practice	Any useful local label
Username	HR	Use the account provided to you
Hostname	db.example.edu	Replace with the real database host

Field	Example value	Notes
Port	1521	Use the port provided by your administrator
Service name	orclpdb.example	Illustrative only—use your actual service name

The values above are examples, not real credentials or a working server. Never share a database password in screenshots or public course materials.

6. Verify the connection

After connecting, open a SQL Worksheet for that connection and run:

```
SELECT SYSDATE FROM dual;
```

Click the Run Statement button (often a green arrow) or press `Ctrl+Enter` on Windows. On macOS, use the keyboard shortcut shown in the SQL Developer menu for your version. A successful result should display the database server's current date and time.

You can also verify the active user with:

```
SELECT USER FROM dual;
```

7. Troubleshooting

Symptom	What to check
Test fails / connection refused	Check hostname, port, VPN/network access, firewall rules, and whether the database endpoint is reachable.
ORA-01017: invalid username/password	Check spelling, password case, and whether the account is locked or the password has expired. Do not repeatedly guess passwords.
ORA-12154: could not resolve connect identifier	For a TNS connection, confirm the alias and that SQL Developer is pointing to the correct network configuration / wallet files.
ORA-12514: listener does not currently know service requested	Confirm the service name with your administrator. A SID and a service name are not interchangeable.
SQL Developer does not launch	Confirm that the archive was fully extracted and that you are launching the executable/app from the extracted folder. If using a package without Java, install the JDK version specified by Oracle for that release.
macOS says the app cannot be opened	Verify the download is from Oracle. Follow your organization's macOS security guidance rather than running commands from unknown websites.

8. Security and workshop checklist

■ Download SQL Developer only from Oracle's official website.

■ Use the package appropriate for the computer's operating system and processor.

■ Use a separate, clean extraction folder; do not extract over an existing version or into an Oracle Database ORACLE_HOME folder.

■ Keep database passwords private. Do not save them on shared computers unless explicitly permitted.

■ If using Autonomous Database, distribute only the wallet and connection details authorized for participants; follow the database administrator's handling instructions.

■ Test the connection before the workshop and run `SELECT SYSDATE FROM dual;` to confirm that it works.

Official references

• Oracle SQL Developer Downloads

• Oracle SQL Developer Installation Guide

• Oracle Windows Installation Notes

Prepared as a practical learner guide. Package names, screenshots, and UI labels may change between releases; Oracle's current download page and release-specific documentation take precedence.